Lab 1: Descriptive Statistics in R

AUTHOR
Dr. Shonn Cheng

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Welcome to Statistics in Education!

This lab covers the following topics:

- Data Structure
- Frequency Tables
- Central Tendency and Variability
- Graphic Summaries
- APA-Style Reporting

Load Packages

```
#install.packages("tidyverse")
#install.packages("haven")
#install.packages("psych")
#install.packages("writexl")
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   3.5.2     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.4
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(haven)
library(psych)
```

```
Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

    %+%, alpha
```

```
library(writexl)
```

Data Structure

Using the `summersurvey.sav` dataset (imported into R as a data frame)

Summer Break Survey

Please check the categories that describe you.
Gender: ____ female ____ male
Degree sought: ____ Master's ____ Ph.D. ____ other, specify
_____
Goal for Summer break (Please select one)
____ Catch up on Work
____ Just Relax
____ Take a Mini-vacation
____ Visit Family and Friends
4. For Summer break, record what your ideal daily temperature would be in degrees Fahrenheit. _____
5. For Summer break, record (in dollars and pennies) what you believe you will pay for a gallon of gasoline for your automobile. _____
6. For Summer break, how much money do you anticipate to spend throughout your summer break given your goal selected in Item 3.
_____
7. I feel that I will have a great Summer break.
____ Disagree
____ Uncertain
____ Agree
Please check the response that best applies to how you feel about statistics
8. I am excited about learning about statistics.
____ Disagree
____ Uncertain
____ Agree
9. I am anxious about taking this statistics class
____ Disagree
____ Uncertain
____ Agree

Survey Codes

Gender: 0=female, 1=male
Degree sought: 0=Master's, 1=PhD, 2=others
Goal for summer break:
1=Catch up on work
2=Just Relax
3=Take a Mini-vacation
4=Visit Family and Friends
I feel that i will have a great summer break: 1=disagree, 2=uncertain, 3=agree
I am excited about learning about statistics: 1=disagree, 2=uncertain, 3=agree
I am anxious about taking this statistics class: 1=disagree, 2=uncertain, 3=agree

```r
#load data
summer <- read_sav("summersurvey.sav")

#check the data structure
summer
```

```
# A tibble: 13 × 10
   gender Degree Degree_Other    Goal  Temp   Gas Money Attitude Excited Anxious
    <dbl>  <dbl> <chr>          <dbl> <dbl> <dbl> <dbl>    <dbl>   <dbl>   <dbl>
 1      1      0 ""                 4    85  0     1500        2       2       3
 2      0      1 ""                 1    75  3.97   200        3       3       1
 3      0      0 ""                 1    77  3.3    500        3       2       2
 4      0      1 ""                 4    75  3.8   1500        3       3       3
 5      0      1 ""                 1    81  3.79  1000        3       2       3
 6      1      1 ""                 1    78  3.87  1200        2       1       3
 7      1      1 ""                 2    76  4.02   799        3       3       3
 8      0      2 ""                 1    78  3.98   500        2       2       3
 9      0      1 ""                 1    70  3.85   400        3       3       3
10      0      0 ""                 3    80  3.7   1500        3       3       1
11      1      0 ""                 1    90  4       NA        3       2       3
12      1      2 "Graduate non…     2    70  3.8    600        3       3       2
13      1      1 ""                 3    82  3.79   500        3       3       1
```

```r
glimpse(summer)
```

```
Rows: 13
Columns: 10
$ gender       <dbl> 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 1, 1
$ Degree       <dbl> 0, 1, 0, 1, 1, 1, 1, 2, 1, 0, 0, 2, 1
$ Degree_Other <chr> "", "", "", "", "", "", "", "", "", "", "", "Graduate non…
$ Goal         <dbl> 4, 1, 1, 4, 1, 1, 2, 1, 1, 3, 1, 2, 3
$ Temp         <dbl> 85, 75, 77, 75, 81, 78, 76, 78, 70, 80, 90, 70, 82
$ Gas          <dbl> 0.00, 3.97, 3.30, 3.80, 3.79, 3.87, 4.02, 3.98, 3.85, 3.7…
$ Money        <dbl> 1500, 200, 500, 1500, 1000, 1200, 799, 500, 400, 1500, NA…
$ Attitude     <dbl> 2, 3, 3, 3, 3, 2, 3, 2, 3, 3, 3, 3, 3
$ Excited      <dbl> 2, 3, 2, 3, 2, 1, 3, 2, 3, 3, 2, 3, 3
$ Anxious      <dbl> 3, 1, 2, 3, 3, 3, 3, 3, 3, 1, 3, 2, 1
```

```r
str(summer)
```

```
tibble [13 × 10] (S3: tbl_df/tbl/data.frame)
 $ gender      : num [1:13] 1 0 0 0 0 1 1 0 0 0 ...
  ..- attr(*, "format.spss")= chr "F1.0"
  ..- attr(*, "display_width")= int 12
 $ Degree      : num [1:13] 0 1 0 1 1 1 1 2 1 0 ...
  ..- attr(*, "format.spss")= chr "F1.0"
  ..- attr(*, "display_width")= int 12
 $ Degree_Other: chr [1:13] "" "" "" "" ...
```

```
 ..- attr(*, "format.spss")= chr "A18"
 ..- attr(*, "display_width")= int 18
$ Goal        : num [1:13] 4 1 1 4 1 1 2 1 1 3 ...
 ..- attr(*, "format.spss")= chr "F1.0"
 ..- attr(*, "display_width")= int 12
$ Temp        : num [1:13] 85 75 77 75 81 78 76 78 70 80 ...
 ..- attr(*, "format.spss")= chr "F2.0"
 ..- attr(*, "display_width")= int 12
$ Gas         : num [1:13] 0 3.97 3.3 3.8 3.79 3.87 4.02 3.98 3.85 3.7 ...
 ..- attr(*, "format.spss")= chr "F17.15"
 ..- attr(*, "display_width")= int 12
$ Money       : num [1:13] 1500 200 500 1500 1000 1200 799 500 400 1500 ...
 ..- attr(*, "format.spss")= chr "F4.0"
 ..- attr(*, "display_width")= int 12
$ Attitude    : num [1:13] 2 3 3 3 3 2 3 2 3 3 ...
 ..- attr(*, "format.spss")= chr "F1.0"
 ..- attr(*, "display_width")= int 12
$ Excited     : num [1:13] 2 3 2 3 2 1 3 2 3 3 ...
 ..- attr(*, "format.spss")= chr "F1.0"
 ..- attr(*, "display_width")= int 12
$ Anxious     : num [1:13] 3 1 2 3 3 3 3 3 3 1 ...
 ..- attr(*, "format.spss")= chr "F1.0"
 ..- attr(*, "display_width")= int 12
```

```r
#change variable type
summer<-summer %>%
  mutate(gender = as.factor(gender))

glimpse(summer)
```

```
Rows: 13
Columns: 10
$ gender       <fct> 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 1, 1
$ Degree       <dbl> 0, 1, 0, 1, 1, 1, 1, 2, 1, 0, 0, 2, 1
$ Degree_Other <chr> "", "", "", "", "", "", "", "", "", "", "", "Graduate non…
$ Goal         <dbl> 4, 1, 1, 4, 1, 1, 2, 1, 1, 3, 1, 2, 3
$ Temp         <dbl> 85, 75, 77, 75, 81, 78, 76, 78, 70, 80, 90, 70, 82
$ Gas          <dbl> 0.00, 3.97, 3.30, 3.80, 3.79, 3.87, 4.02, 3.98, 3.85, 3.7…
$ Money        <dbl> 1500, 200, 500, 1500, 1000, 1200, 799, 500, 400, 1500, NA…
$ Attitude     <dbl> 2, 3, 3, 3, 3, 2, 3, 2, 3, 3, 3, 3, 3
$ Excited      <dbl> 2, 3, 2, 3, 2, 1, 3, 2, 3, 3, 2, 3, 3
$ Anxious      <dbl> 3, 1, 2, 3, 3, 3, 3, 3, 3, 1, 3, 2, 1
```

Are there any other variables whose variable type need to be changed?

```r
#Degree

summer<-summer %>%
  mutate(Degree = as.factor(Degree))

#Goal

summer<-summer %>%
  mutate(Goal = as.factor(Goal))

summer
```

```
# A tibble: 13 × 10
   gender Degree Degree_Other   Goal   Temp   Gas Money Attitude Excited Anxious
   <fct>  <fct>  <chr>         <fct> <dbl> <dbl> <dbl>    <dbl>   <dbl>   <dbl>
 1 1      0      ""            4        85  0     1500        2       2       3
 2 0      1      ""            1        75  3.97   200        3       3       1
 3 0      0      ""            1        77  3.3    500        3       2       2
 4 0      1      ""            4        75  3.8   1500        3       3       3
 5 0      1      ""            1        81  3.79  1000        3       2       3
 6 1      1      ""            1        78  3.87  1200        2       1       3
 7 1      1      ""            2        76  4.02   799        3       3       3
 8 0      2      ""            1        78  3.98   500        2       2       3
 9 0      1      ""            1        70  3.85   400        3       3       3
10 0      0      ""            3        80  3.7   1500        3       3       1
11 1      0      ""            1        90  4       NA        3       2       3
12 1      2      "Graduate non… 2       70  3.8    600        3       3       2
13 1      1      ""            3        82  3.79   500        3       3       1
```

Frequency Table

A frequency table takes a set of scores and tabulates them in order from highest to lowest (or lowest to highest), grouping together all individuals who have the same (or same range of) score(s).

Get Frequency Tables in R

To create a frequency table in R, use the `table()` function:

```
# Example: Frequency table for a categorical variable

summer
```

```
# A tibble: 13 × 10
   gender Degree Degree_Other   Goal   Temp   Gas Money Attitude Excited Anxious
   <fct>  <fct>  <chr>          <fct> <dbl> <dbl> <dbl>    <dbl>   <dbl>   <dbl>
 1 1      0      ""             4        85  0     1500        2       2       3
 2 0      1      ""             1        75  3.97   200        3       3       1
 3 0      0      ""             1        77  3.3    500        3       2       2
 4 0      1      ""             4        75  3.8   1500        3       3       3
 5 0      1      ""             1        81  3.79  1000        3       2       3
 6 1      1      ""             1        78  3.87  1200        2       1       3
 7 1      1      ""             2        76  4.02   799        3       3       3
 8 0      2      ""             1        78  3.98   500        2       2       3
 9 0      1      ""             1        70  3.85   400        3       3       3
10 0      0      ""             3        80  3.7   1500        3       3       1
11 1      0      ""             1        90  4       NA        3       2       3
12 1      2      "Graduate non… 2        70  3.8    600        3       3       2
13 1      1      ""             3        82  3.79   500        3       3       1
```

```
table(summer$gender)
```

```
0 1
7 6
```

What percentage of students plan to catch up work in the summer break? What percentage plan to “just relax”?

```
#frequency table
table(summer$Goal)
```

```
1 2 3 4
7 2 2 2
```

```
#solution 1
prop.table(table(summer$Goal))*100
```

```
       1        2        3        4
53.84615 15.38462 15.38462 15.38462
```

```
#solution 2
summer %>%
  group_by(Goal) %>%
  summarize(n = n()) %>%
  mutate(percent = n/13*100)
```

```
# A tibble: 4 × 3
  Goal      n percent
  <fct> <int>   <dbl>
1 1         7    53.8
2 2         2    15.4
3 3         2    15.4
4 4         2    15.4
```

Practice

Using the `lab 1 example data.sav` dataset (imported into R as a data frame):

- gender(1=Male;2=Female),
- race(1=Caucasian; 2=African American; 3=Hispanic; 4=Other),
- scitype(1=Social; 2=Physical),
- stutype(1=Domestic; 2=Foreign)

1. Check the data structure. Are there any variables whose variable type need to be changed?
2. How many students study social sciences? How many study physical sciences?
3. List all the possible values of College GPA in descending order. What is the highest score? What percentage of students got that score?

```r
#load data
lab1 <- read_sav("lab 1 example data.sav")

lab1
```

```
# A tibble: 44 × 10
      id gender  race   age scitype stutype  grev  greq  cgpa  ggpa
   <dbl>  <dbl> <dbl> <dbl>   <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
 1     1      1     1    24       1       1   450   500  3.14  3.42
 2    24      1     2    24       1       1   420   590  3.15  3.43
 3    37      1     3    24       2       1   490   550  3.31  3.5
 4     2      1     1    25       1       1   480   580  3.11  3.34
 5     3      1     1    30       1       1   610   650  3.62  3.79
 6     4      1     1    30       1       1   540   550  3.15  3.46
 7     9      1     1    27       1       2   500   640  3.24  3.57
 8    10      1     1    30       1       2   490   740  3.26  3.48
 9    25      1     2    28       1       1   580   590  3.54  3.78
10    26      1     2    29       1       1   620   650  3.54  3.85
# ℹ 34 more rows
```

```r
glimpse(lab1)
```

```
Rows: 44
Columns: 10
$ id      <dbl> 1, 24, 37, 2, 3, 4, 9, 10, 25, 26, 28, 13, 21, 23, 29, 38, 43,…
$ gender  <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,…
$ race    <dbl> 1, 2, 3, 1, 1, 1, 1, 1, 2, 2, 2, 1, 1, 1, 2, 3, 4, 1, 1, 1, 1,…
$ age     <dbl> 24, 24, 24, 25, 30, 30, 27, 30, 28, 29, 29, 26, 28, 29, 27, 29…
$ scitype <dbl> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,…
$ stutype <dbl> 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 2, 1, 2, 2, 1, 1, 1, 1, 1, 1, 2,…
$ grev    <dbl> 450, 420, 490, 480, 610, 540, 500, 490, 580, 620, 640, 500, 42…
$ greq    <dbl> 500, 590, 550, 580, 650, 550, 640, 740, 590, 650, 660, 480, 75…
$ cgpa    <dbl> 3.14, 3.15, 3.31, 3.11, 3.62, 3.15, 3.24, 3.26, 3.54, 3.54, 3.…
$ ggpa    <dbl> 3.42, 3.43, 3.50, 3.34, 3.79, 3.46, 3.57, 3.48, 3.78, 3.85, 3.…
```

```r
#change varible type
lab1<-lab1 %>% mutate(gender = as.factor(gender),
                race = as.factor(race),
                scitype = as.factor(scitype),
                stutype = as.factor(stutype)
                )

#example: frequency table for major
#scitype(1=Social; 2=Physical)

table(lab1$scitype)
```

```
 1  2
21 23
```

```r
#example: frequency table for college GPA
table(lab1$cgpa)
```

```
3.11 3.14 3.15 3.18  3.2 3.21 3.22 3.24 3.26 3.28 3.29 3.31 3.33 3.34 3.37 3.38
   1    1    2    1    1    1    1    1    2    1    1    1    1    2    1    1
3.39  3.4 3.42 3.45 3.46 3.47  3.5 3.54 3.56 3.58 3.62 3.64 3.68 3.75
   2    2    1    4    2    1    2    3    1    1    2    2    1    1
```

```r
#descending order
lab1 %>%
  count(cgpa) %>%
  arrange(desc(cgpa))
```

```
# A tibble: 30 × 2
    cgpa     n
   <dbl> <int>
 1  3.75     1
 2  3.68     1
 3  3.64     2
 4  3.62     2
 5  3.58     1
 6  3.56     1
 7  3.54     3
 8  3.5      2
 9  3.47     1
10  3.46     2
# ℹ 20 more rows
```

```r
#percentage
lab1 %>%
  group_by(cgpa) %>%
  summarize(n = n()) %>%
  arrange(desc(cgpa)) %>%
  mutate(percent = n/44*100)
```

```
# A tibble: 30 × 3
    cgpa     n percent
   <dbl> <int>   <dbl>
 1  3.75     1    2.27
 2  3.68     1    2.27
 3  3.64     2    4.55
 4  3.62     2    4.55
 5  3.58     1    2.27
 6  3.56     1    2.27
 7  3.54     3    6.82
 8  3.5      2    4.55
 9  3.47     1    2.27
10  3.46     2    4.55
# ℹ 20 more rows
```

Descriptive Statistics

Descriptive statistics provide a quantitative description of the main features of a collection of data. Examples include measures of central tendency and variability.

Central Tendency and Variability

- **Central Tendency**: Mean, Median, Mode
- **Variability**: Range, Interquartile Range (IQR), Variance, Standard Deviation (SD)

Measures of Variability

- **Range**: `max - min`
- **Interquartile Range (IQR)**: `Q3 - Q1` (middle 50% of the distribution)
- **Variance & Standard Deviation**: Measures of average deviation from the mean.

```r
#solution1
summary(summer)
```

```
 gender Degree Degree_Other       Goal       Temp            Gas       
 0:7    0:4    Length:13          1:7   Min.   :70.00   Min.   :0.000  
 1:6    1:7    Class :character   2:2   1st Qu.:75.00   1st Qu.:3.790  
        2:2    Mode  :character   3:2   Median :78.00   Median :3.800  
                                  4:2   Mean   :78.23   Mean   :3.528  
                                        3rd Qu.:81.00   3rd Qu.:3.970  
                                        Max.   :90.00   Max.   :4.020  

     Money           Attitude        Excited         Anxious      
 Min.   : 200.0   Min.   :2.000   Min.   :1.000   Min.   :1.000  
 1st Qu.: 500.0   1st Qu.:3.000   1st Qu.:2.000   1st Qu.:2.000  
 Median : 699.5   Median :3.000   Median :3.000   Median :3.000  
 Mean   : 849.9   Mean   :2.769   Mean   :2.462   Mean   :2.385  
 3rd Qu.:1275.0   3rd Qu.:3.000   3rd Qu.:3.000   3rd Qu.:3.000  
 Max.   :1500.0   Max.   :3.000   Max.   :3.000   Max.   :3.000  
 NA's   :1                                                       
```

```r
#solution2
describe(summer)
```

```
              vars  n   mean     sd median trimmed    mad min     max   range
gender*          1 13   1.46   0.52    1.0    1.45   0.00   1    2.00    1.00
Degree*          2 13   1.85   0.69    2.0    1.82   0.00   1    3.00    2.00
Degree_Other*    3 13   1.08   0.28    1.0    1.00   0.00   1    2.00    1.00
Goal*            4 13   1.92   1.19    1.0    1.82   0.00   1    4.00    3.00
Temp             5 13  78.23   5.57   78.0   77.91   4.45  70   90.00   20.00
Gas              6 13   3.53   1.08    3.8    3.80   0.15   0    4.02    4.02
Money            7 12 849.92 473.87  699.5  849.90 444.78 200 1500.00 1300.00
Attitude         8 13   2.77   0.44    3.0    2.82   0.00   2    3.00    1.00
Excited          9 13   2.46   0.66    3.0    2.55   0.00   1    3.00    2.00
Anxious         10 13   2.38   0.87    3.0    2.45   0.00   1    3.00    2.00
               skew kurtosis     se
gender*        0.14    -2.13   0.14
Degree*        0.16    -1.09   0.19
```

```
Degree_Other*  2.82     6.44   0.08
Goal*          0.68    -1.26   0.33
Temp           0.37    -0.52   1.54
Gas           -2.68     5.90   0.30
Money          0.28    -1.65 136.80
Attitude      -1.13    -0.76   0.12
Excited       -0.67    -0.82   0.18
Anxious       -0.73    -1.36   0.24
```

Practice

Using the `lab 1 example data.sav` dataset:

1. Calculate the mean, median, and standard deviation of age, grev, and greq.

```r
summary(lab1)
```

```
       id        gender race        age        scitype stutype      grev
 Min.   : 1.00   1:26   1:23   Min.   :23.00   1:21    1:31    Min.   :390.0
 1st Qu.:11.75   2:18   2:11   1st Qu.:26.00   2:23    2:13    1st Qu.:497.5
 Median :22.50          3: 6   Median :29.00                   Median :580.0
 Mean   :22.50          4: 4   Mean   :29.02                   Mean   :558.4
 3rd Qu.:33.25                 3rd Qu.:32.00                   3rd Qu.:622.5
 Max.   :44.00                 Max.   :36.00                   Max.   :700.0
      greq            cgpa            ggpa
 Min.   :480.0   Min.   :3.110   Min.   :3.310
 1st Qu.:540.0   1st Qu.:3.275   1st Qu.:3.487
 Median :575.0   Median :3.400   Median :3.620
 Mean   :596.4   Mean   :3.401   Mean   :3.627
 3rd Qu.:650.0   3rd Qu.:3.510   3rd Qu.:3.772
 Max.   :750.0   Max.   :3.750   Max.   :3.900
```

```r
describe(lab1)
```

```
         vars  n   mean    sd median trimmed   mad    min    max  range  skew
id          1 44  22.50 12.85  22.50   22.50 16.31   1.00  44.00  43.00  0.00
gender*     2 44   1.41  0.50   1.00    1.39  0.00   1.00   2.00   1.00  0.36
race*       3 44   1.80  1.00   1.00    1.64  0.00   1.00   4.00   3.00  0.95
age         4 44  29.02  3.35  29.00   28.97  4.45  23.00  36.00  13.00  0.02
scitype*    5 44   1.52  0.51   2.00    1.53  0.00   1.00   2.00   1.00 -0.09
stutype*    6 44   1.30  0.46   1.00    1.25  0.00   1.00   2.00   1.00  0.87
grev        7 44 558.41 83.69 580.00  561.67 88.96 390.00 700.00 310.00 -0.31
greq        8 44 596.36 77.16 575.00  591.39 74.13 480.00 750.00 270.00  0.56
cgpa        9 44   3.40  0.16   3.40    3.40  0.19   3.11   3.75   0.64  0.06
ggpa       10 44   3.63  0.17   3.62    3.63  0.21   3.31   3.90   0.59 -0.06
         kurtosis    se
id          -1.28  1.94
gender*     -1.91  0.07
race*       -0.38  0.15
age         -1.02  0.51
scitype*    -2.04  0.08
stutype*    -1.28  0.07
grev        -1.04 12.62
greq        -0.82 11.63
cgpa        -0.88  0.02
ggpa        -1.30  0.02
```

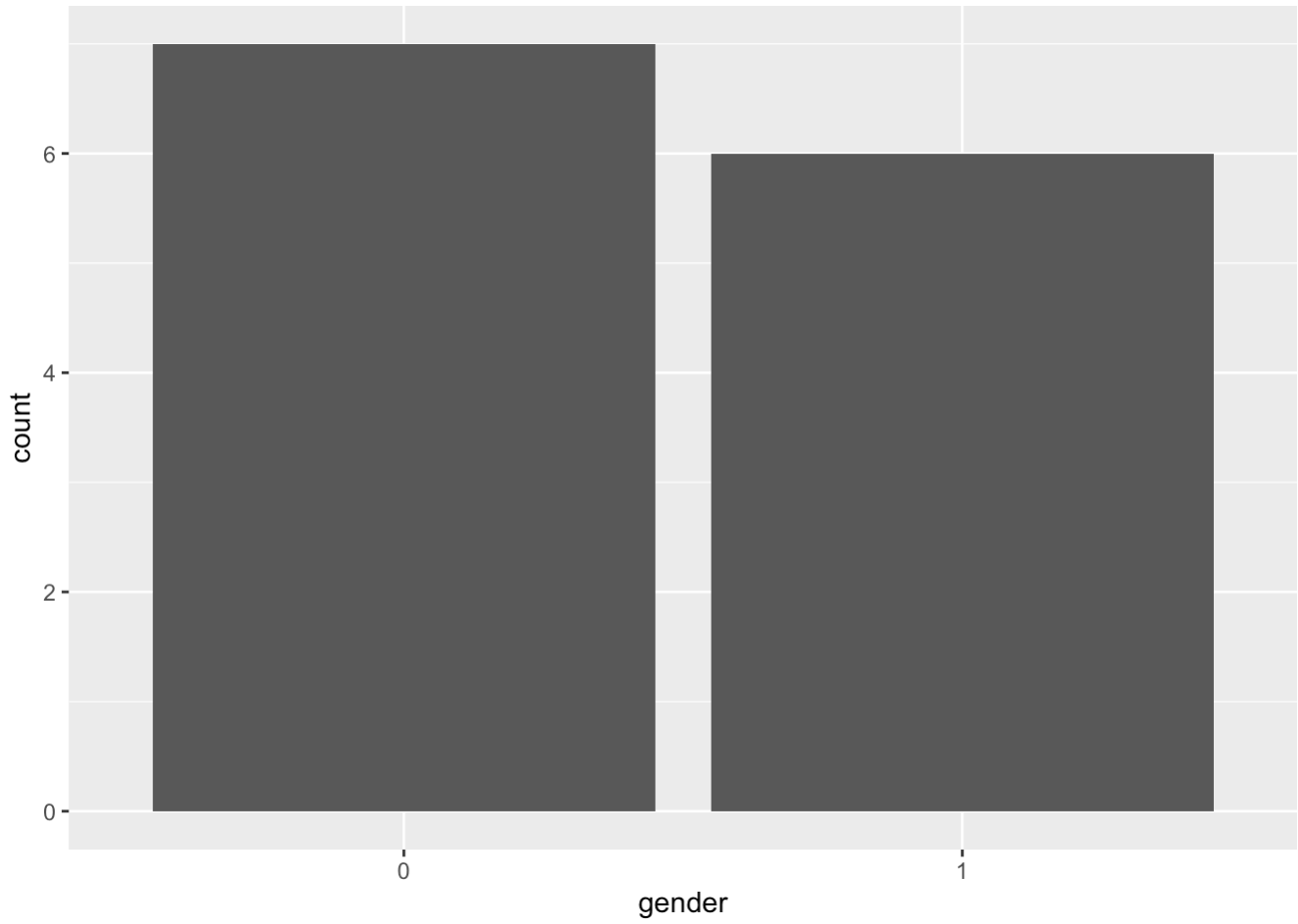
Graphic Summaries

Graphic summaries (charts) are useful for visualizing data distributions.

Creating Charts in R

Bar Charts

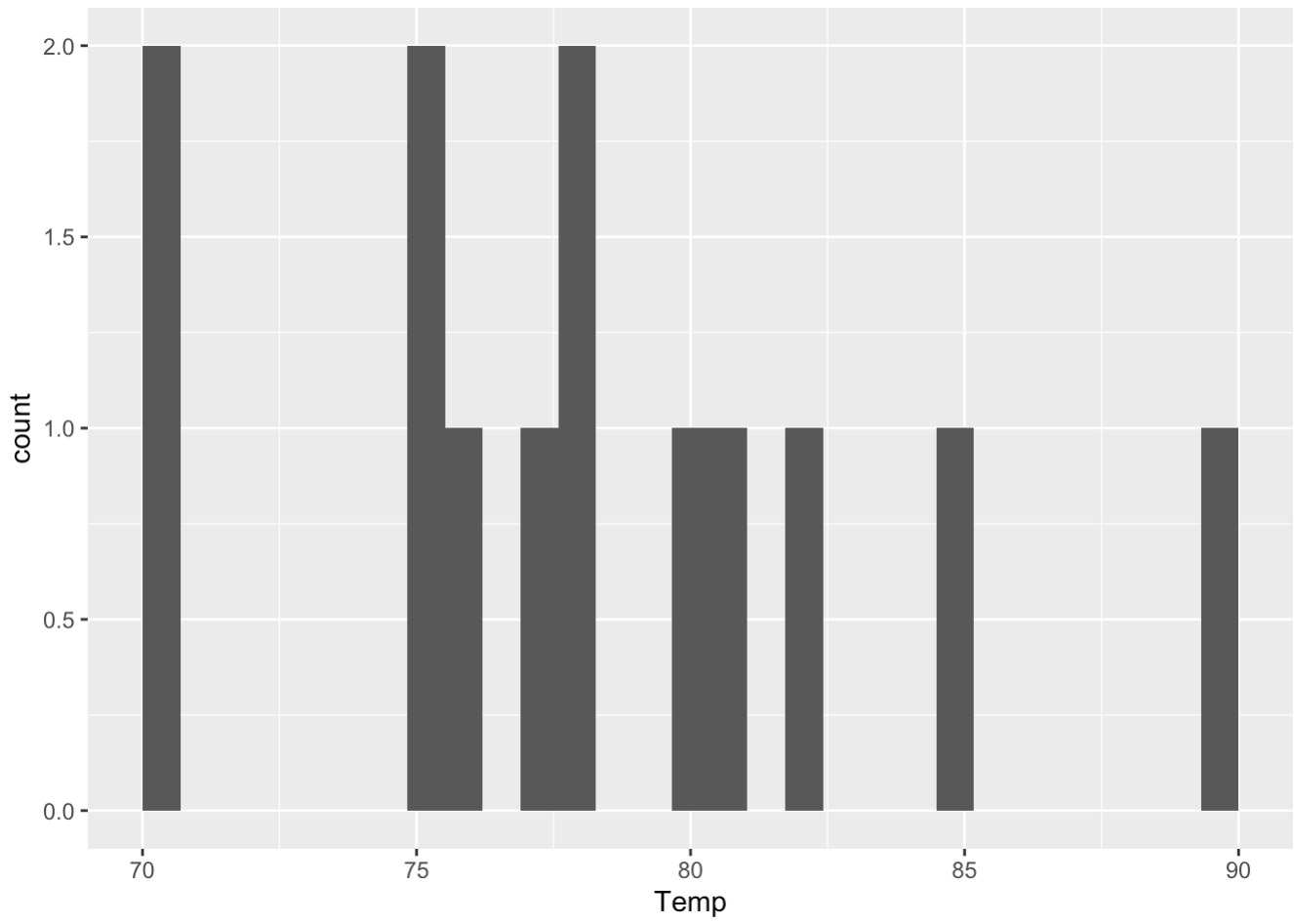
```r
#categorical variable

summer %>%
  ggplot(mapping = aes (x = gender)) +
  geom_bar()
```



Histograms

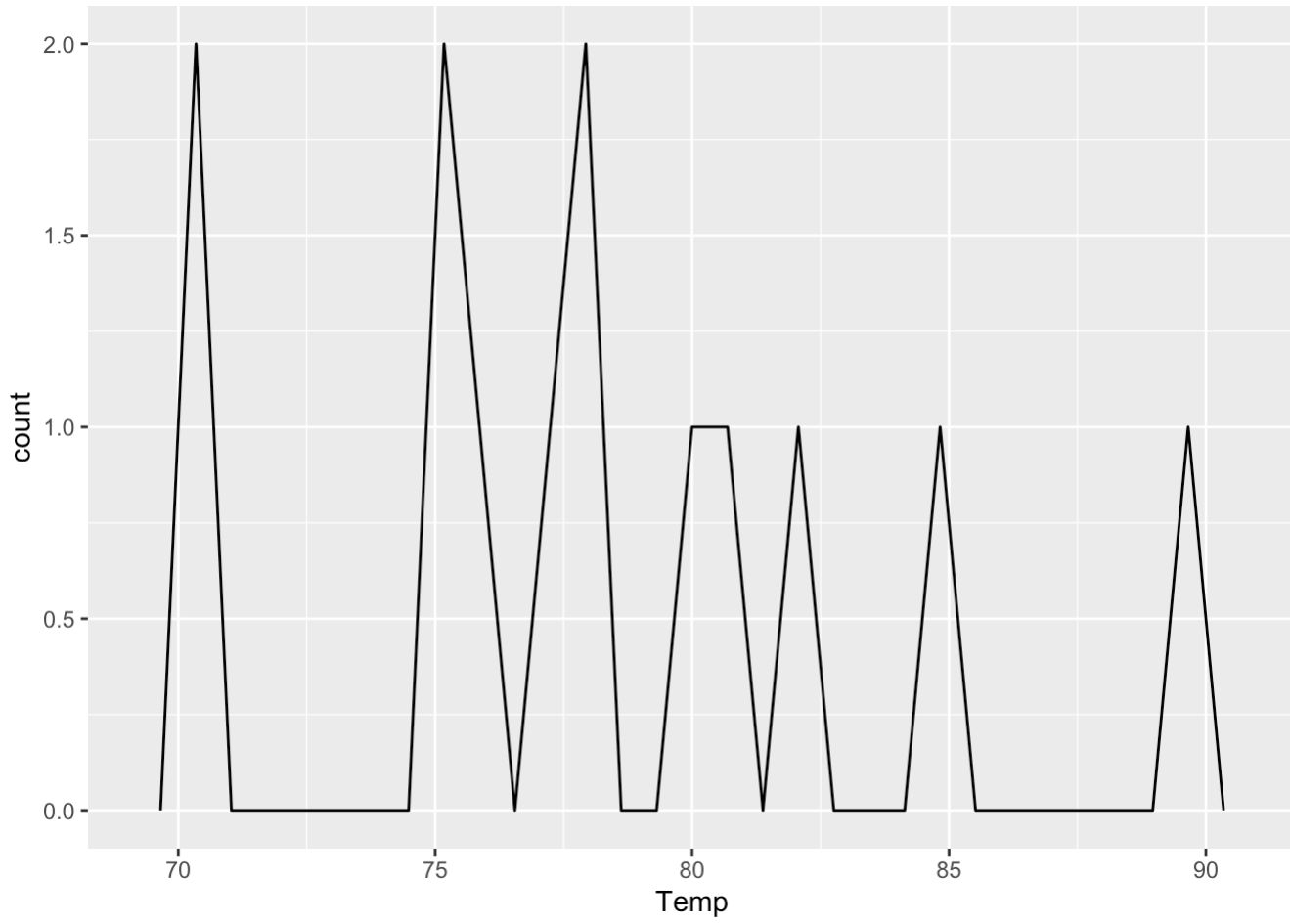
```
#continuous variable

summer %>%
  ggplot(mapping = aes (x = Temp)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

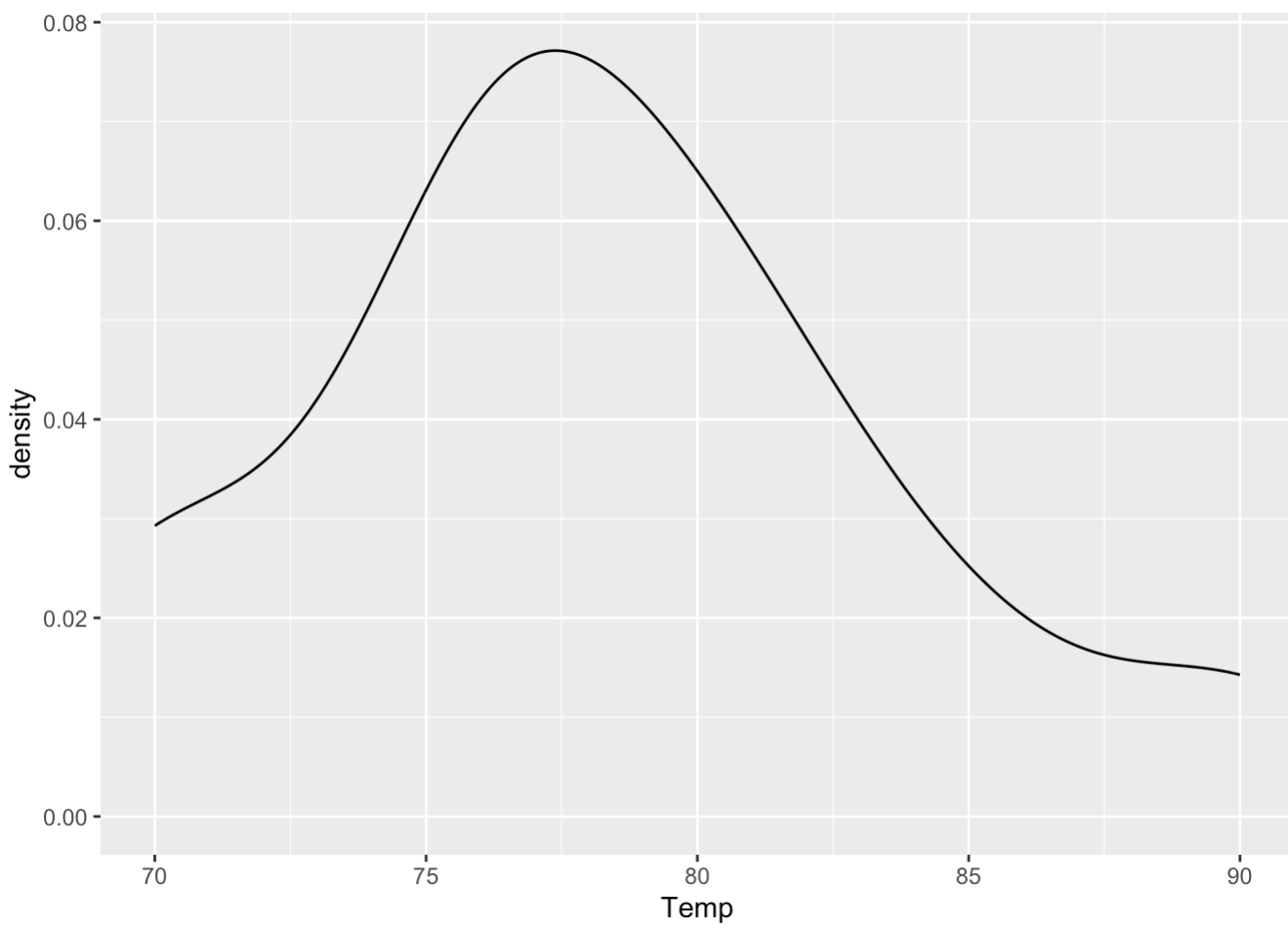



```
summer %>%
  ggplot(mapping = aes (x = Temp)) +
  geom_freqpoly()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

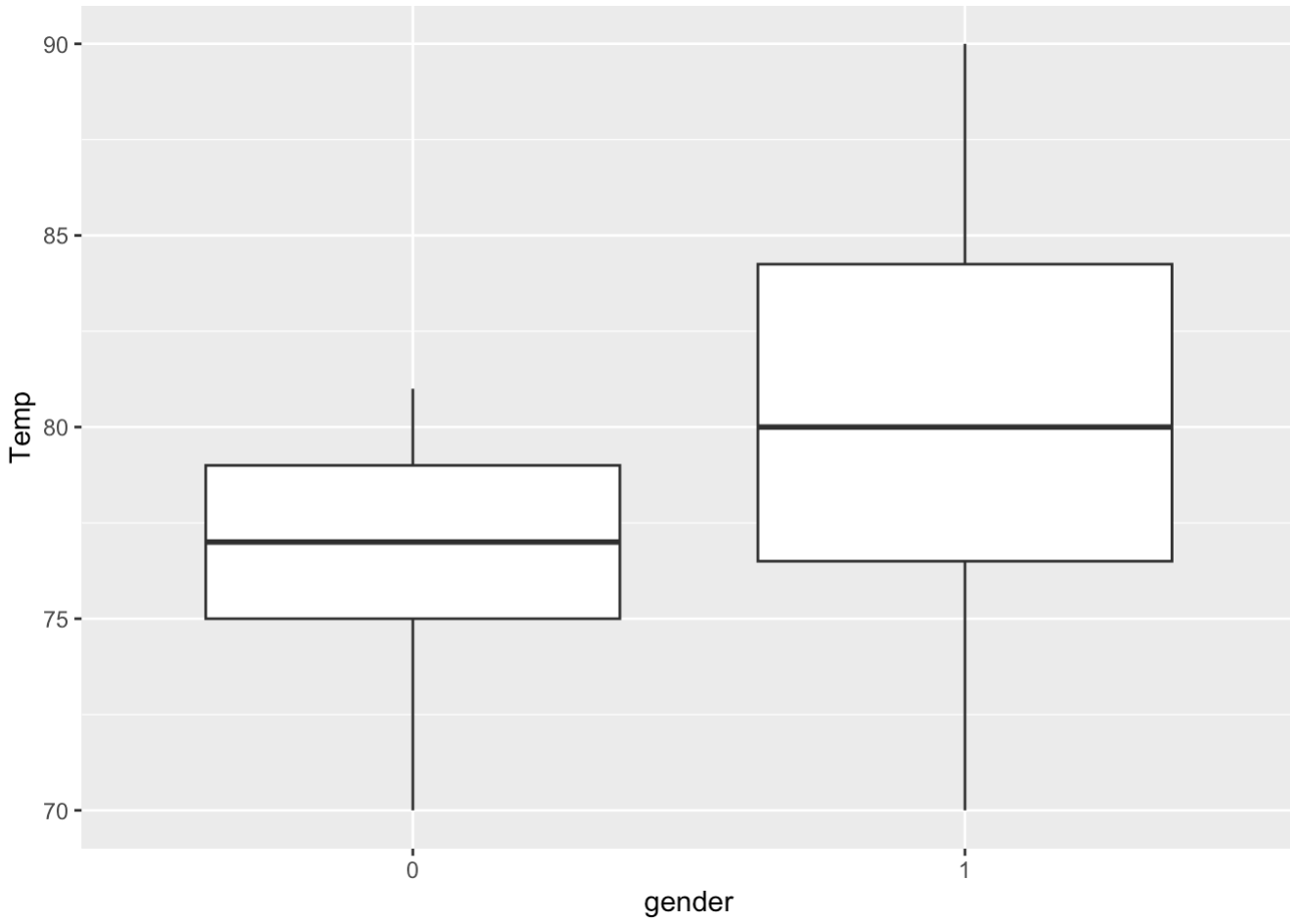



```
summer %>%
  ggplot(mapping = aes (x = Temp)) +
  geom_density()
```



Boxplots

```
#continuous variable by group

summer %>%
  ggplot(mapping = aes (x = gender, y = Temp)) +
  geom_boxplot()
```

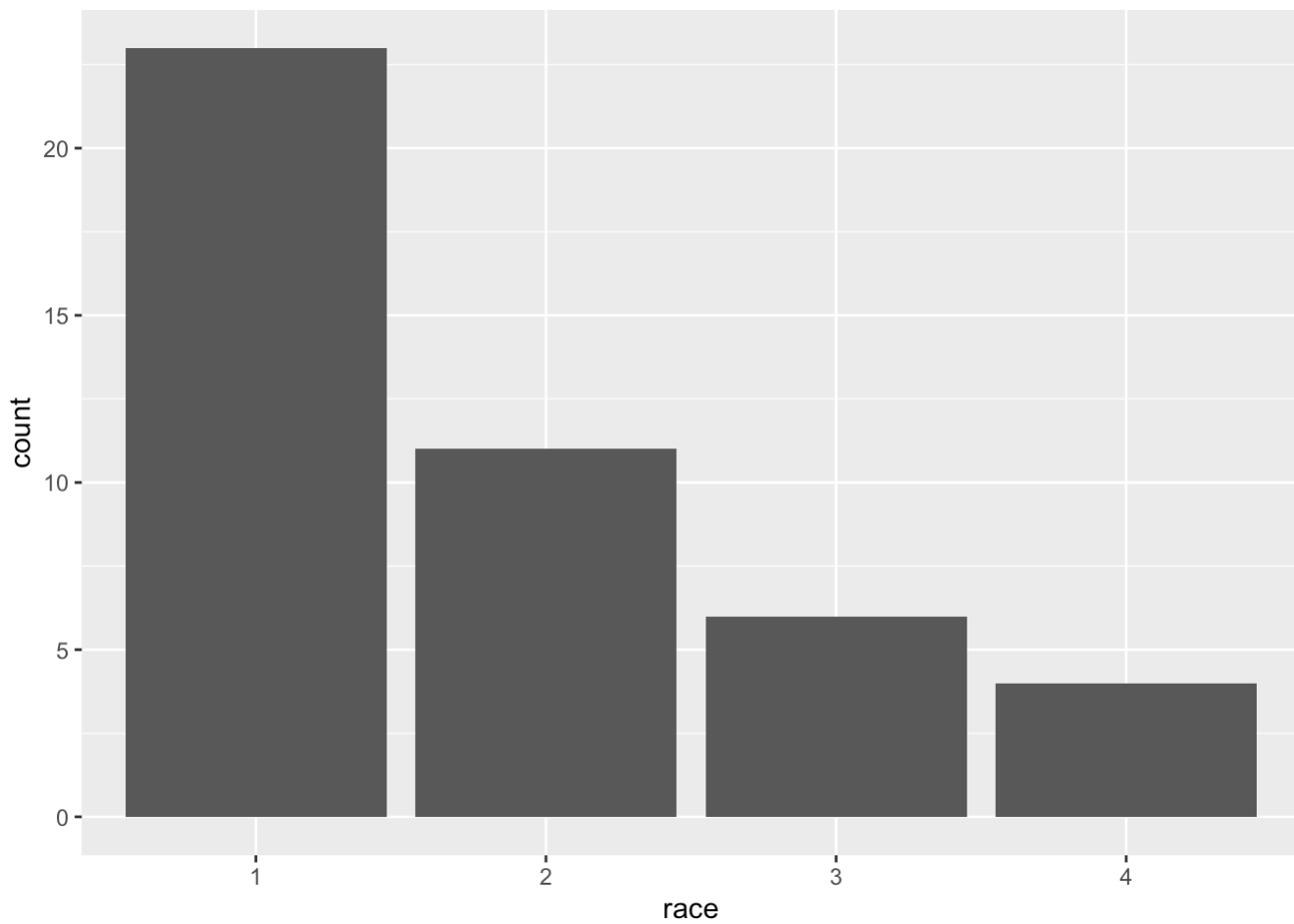


Practice

Using the `lab 1 example data.sav` dataset:

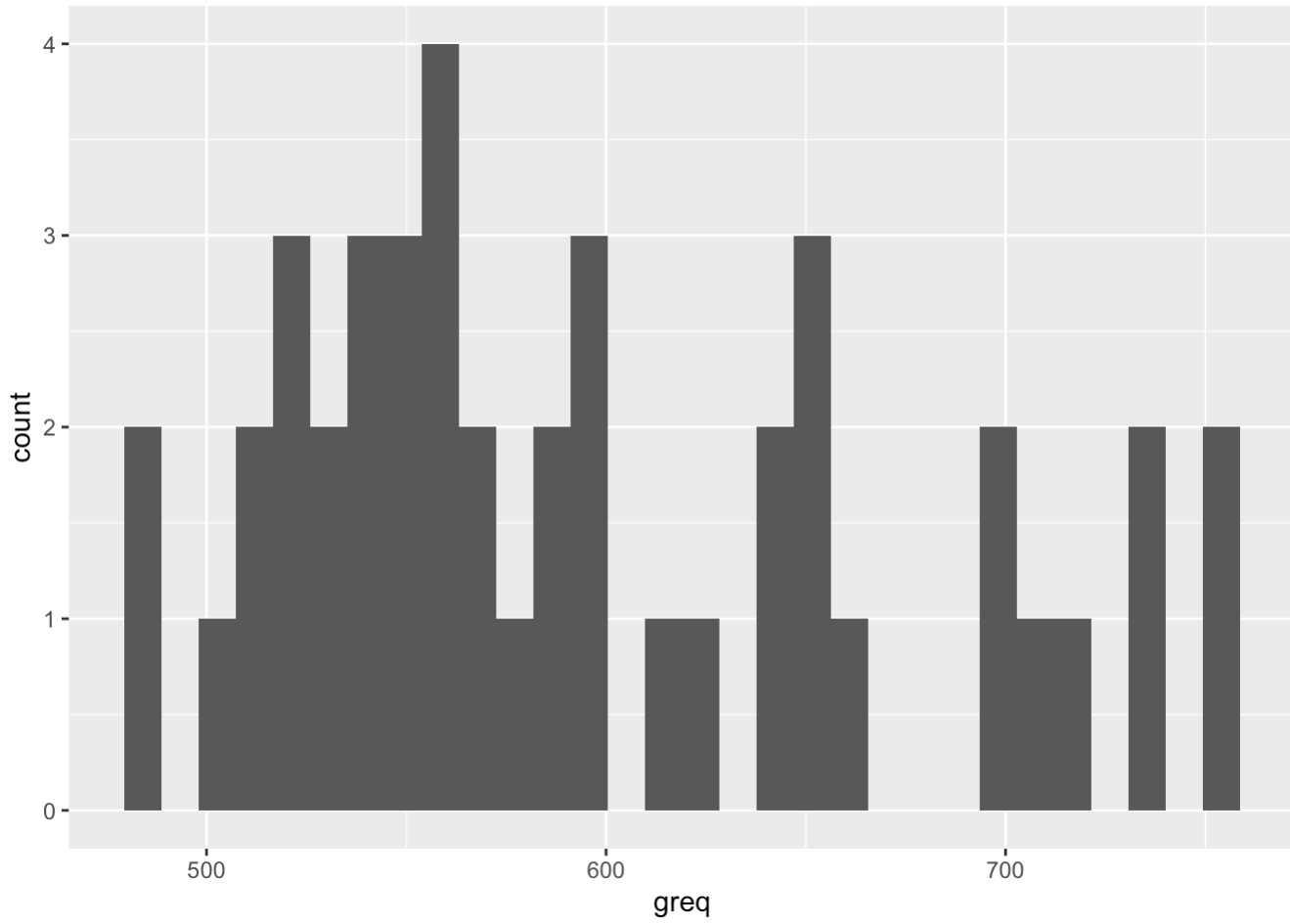
1. Create a bar chart for `race`.
2. Create a histogram for `greq`.
3. Create a boxplot to show `greq` by `race`.

```
#task 1
lab1 %>%
  ggplot(mapping = aes (x = race)) +
  geom_bar()
```

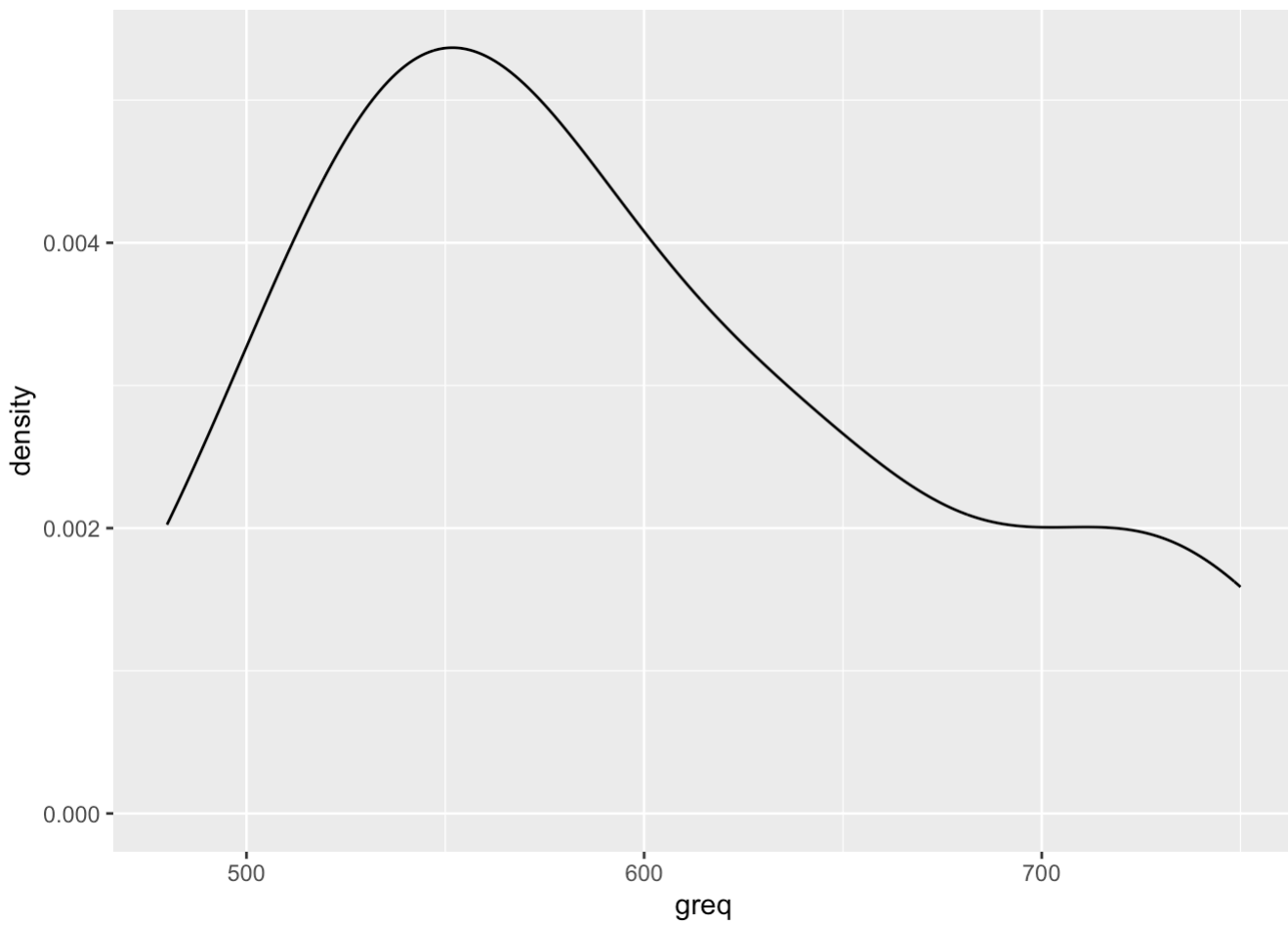



```
#task 2
lab1 %>%
  ggplot(mapping = aes (x = greq)) +
  geom_histogram()
```

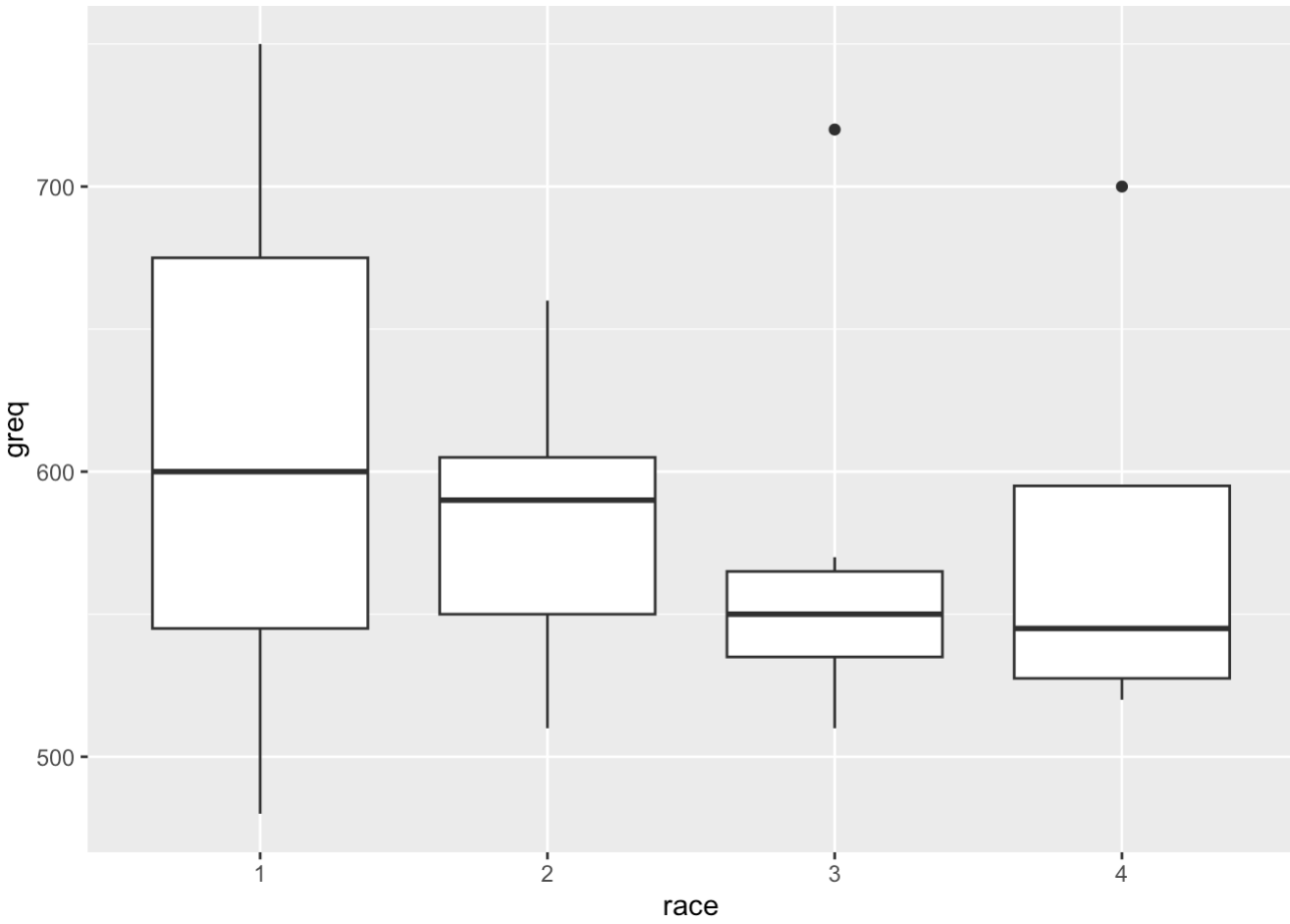
```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
#task 3
lab1 %>%
  ggplot(mapping = aes (x = greq)) +
  geom_density()
```




```
lab1 %>%
  ggplot(mapping = aes (x = race, y = greq)) +
  geom_boxplot()
```



APA-Style Reporting

```
summer %>%
  select(Temp:Anxious) %>%
  describe()
```

```
         vars  n   mean     sd median trimmed    mad min     max   range  skew
Temp        1 13  78.23   5.57   78.0   77.91   4.45  70   90.00   20.00  0.37
Gas         2 13   3.53   1.08    3.8    3.80   0.15   0    4.02    4.02 -2.68
Money       3 12 849.92 473.87  699.5  849.90 444.78 200 1500.00 1300.00  0.28
Attitude    4 13   2.77   0.44    3.0    2.82   0.00   2    3.00    1.00 -1.13
Excited     5 13   2.46   0.66    3.0    2.55   0.00   1    3.00    2.00 -0.67
Anxious     6 13   2.38   0.87    3.0    2.45   0.00   1    3.00    2.00 -0.73
         kurtosis     se
Temp        -0.52   1.54
Gas          5.90   0.30
Money       -1.65 136.80
Attitude    -0.76   0.12
Excited     -0.82   0.18
Anxious     -1.36   0.24
```

```
des<-summer %>%
  select(Temp:Anxious) %>%
  describe()

write_xlsx(des, "des.xlsx")
```

Practice

Using the `lab 1 example data.sav` dataset:

1. Generate descriptive statistics for `age`, `grev`, `greq`, `cgpa`, and `ggpa`.
2. Format the results in an APA-style table, ensuring appropriate labels and formatting.
3. Write a brief summary interpreting the descriptive statistics.

```
data<-read_sav("lab 1 example data.sav")

data
```

```
# A tibble: 44 × 10
      id gender  race   age scitype stutype  grev  greq  cgpa  ggpa
   <dbl>  <dbl> <dbl> <dbl>   <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
 1     1      1     1    24       1       1   450   500  3.14  3.42
 2    24      1     2    24       1       1   420   590  3.15  3.43
 3    37      1     3    24       2       1   490   550  3.31  3.5
 4     2      1     1    25       1       1   480   580  3.11  3.34
 5     3      1     1    30       1       1   610   650  3.62  3.79
 6     4      1     1    30       1       1   540   550  3.15  3.46
 7     9      1     1    27       1       2   500   640  3.24  3.57
```

```
 8    10      1     1    30       1       2   490   740  3.26  3.48
 9    25      1     2    28       1       1   580   590  3.54  3.78
10    26      1     2    29       1       1   620   650  3.54  3.85
# ℹ 34 more rows
```

```
df<-data %>%
  select(age, grev, greq, cgpa, ggpa) %>%
  describe()

write_xlsx(df, "df.xlsx")
```